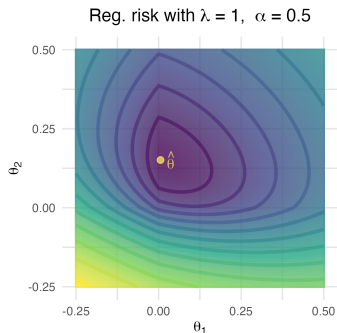
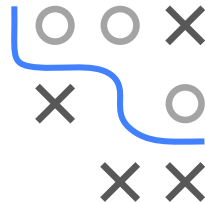


# Optimization in Machine Learning

## Optimization Problems

## Unconstrained problems



### Learning goals

- Definition
- Max. likelihood
- Linear regression
- Regularized risk minimization
- SVM
- Neural network

# UNCONSTRAINED OPTIMIZATION PROBLEM

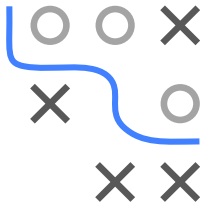
$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$$

with objective function

$$f : \mathcal{S} \rightarrow \mathbb{R}$$

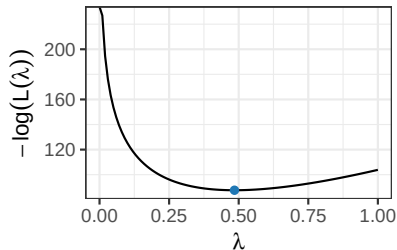
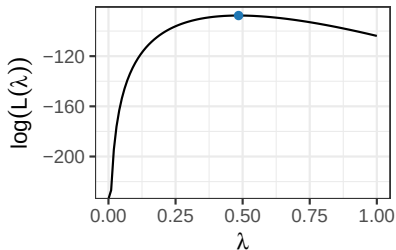
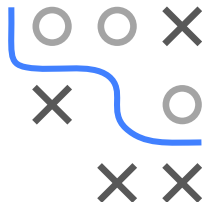
The problem is called

- **unconstrained**, if  $\mathcal{S} = \mathbb{R}^d$
- **smooth** if  $f$  is at least  $\in \mathcal{C}^1$
- **univariate** if  $d = 1$ , and **multivariate** if  $d > 1$
- **convex** if  $f$  convex function (on convex  $\mathbb{R}^d$ )



# NOTE: A CONVENTION IN OPTIMIZATION

- W.l.o.g., we always **minimize** functions  $f$ .
- Maximization is handled by minimizing  $-f$

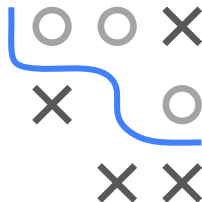
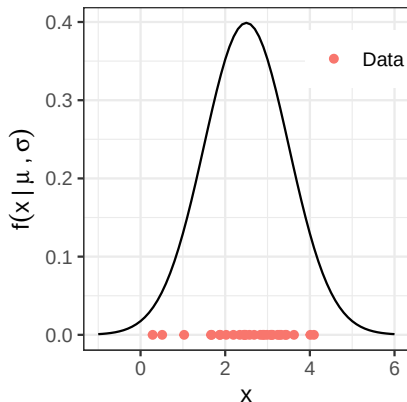


# EXAMPLE 1: MAXIMUM LIKELIHOOD

- $\mathcal{D} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}) \stackrel{\text{i.i.d.}}{\sim} f(\mathbf{x} \mid \mu, \sigma)$  with  $\sigma = 1$ :

$$f(\mathbf{x} \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(\mathbf{x} - \mu)^2}{2\sigma^2}\right)$$

- **Goal:** Find  $\mu \in \mathbb{R}$  which makes observed data most likely



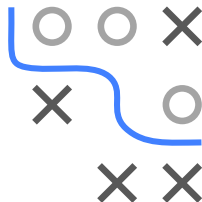
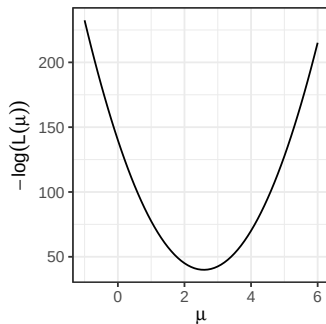
# EXAMPLE 1: MAXIMUM LIKELIHOOD

- **Likelihood:**

$$\mathcal{L}(\mu | \mathcal{D}) = \prod_{i=1}^n f(\mathbf{x}^{(i)} | \mu, 1) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}^{(i)} - \mu)^2\right)$$

- **Neg. log-likelihood:**

$$-\ell(\mu, \mathcal{D}) = -\log \mathcal{L}(\mu | \mathcal{D}) = \frac{n}{2} \log(2\pi) + \frac{1}{2} \sum_{i=1}^n (\mathbf{x}^{(i)} - \mu)^2$$

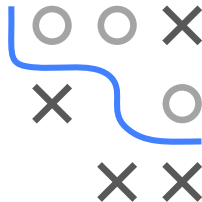
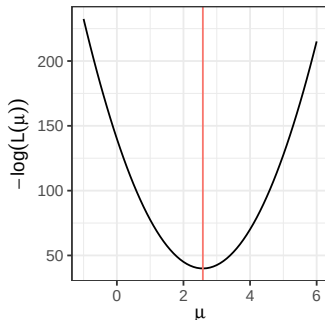


# EXAMPLE 1: MAXIMUM LIKELIHOOD

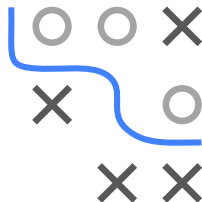
$$\min_{\mu \in \mathbb{R}} -\ell(\mu, \mathcal{D}).$$

- can be solved analytically (setting the first deriv. to 0) since it is a quadratic function:

$$-\frac{\partial \ell(\mu, \mathcal{D})}{\partial \mu} = \sum_{i=1}^n (\mathbf{x}^{(i)} - \mu) = 0 \quad \Leftrightarrow \quad \hat{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)}$$



## EXAMPLE 1: MAXIMUM LIKELIHOOD



- Was: **smooth, univariate, unconstrained, convex**
- If we had optimized for  $\sigma$  as well (instead of assuming it as fixed)

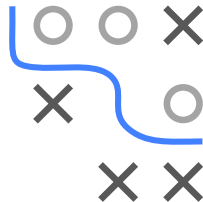
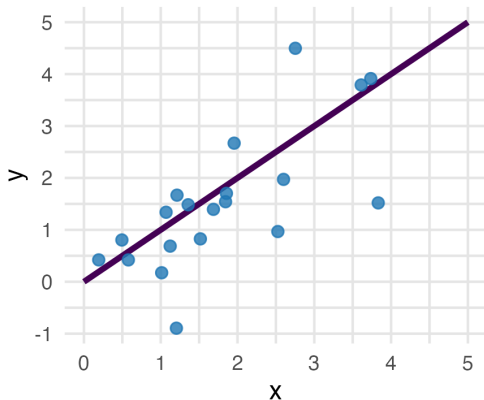
$$\min_{\mu \in \mathbb{R}, \sigma \in \mathbb{R}^+} -\ell(\mu, \mathcal{D})$$

- The problem would have been bivariate and constrained

## EXAMPLE 2: NORMAL REGRESSION

- Assume (multivariate) data  $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$  and we want to fit a linear function to it

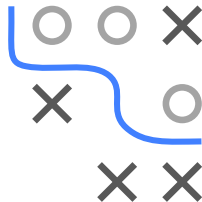
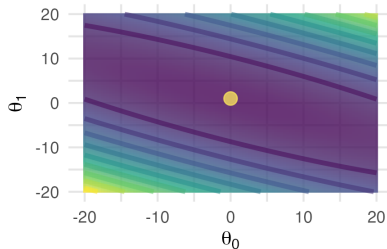
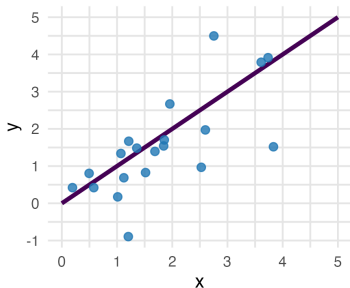
$$y = f(\mathbf{x}) = \boldsymbol{\theta}^\top \mathbf{x}$$



## EXAMPLE 2: LEAST SQUARES LINEAR REGR.

- Find param vector  $\theta$  that minimizes SSE / risk with L2 loss

$$\min_{\theta \in \mathbb{R}^d} \sum_{i=1}^n \left( \theta^\top \mathbf{x}^{(i)} - y^{(i)} \right)^2$$



- Smooth, multivariate, unconstrained, convex** problem
- Quadratic function
- Analytic solution:  $\theta = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ , where  $\mathbf{X}$  is design matrix

# RISK MINIMIZATION IN ML

- In the above example, if we exchange

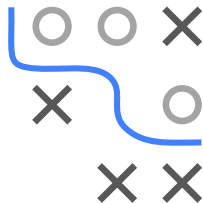
$$\min_{\theta \in \mathbb{R}^d} \sum_{i=1}^n \left( \theta^\top \mathbf{x}^{(i)} - y^{(i)} \right)^2$$

- the linear model  $\theta^\top \mathbf{x}$  by an arbitrary model  $f(\mathbf{x} \mid \theta)$
- the L2-loss  $(f(\mathbf{x} \mid \theta) - y)^2$  by any loss  $L(y, f(\mathbf{x}))$
- we arrive at general **empirical risk minimization** (ERM)

$$\mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^n L\left(y^{(i)}, f(\mathbf{x}^{(i)} \mid \theta)\right) = \min!$$

- Usually, we add a regularizer to counteract overfitting:

$$\mathcal{R}_{\text{reg}}(\theta) = \sum_{i=1}^n L\left(y^{(i)}, f(\mathbf{x}^{(i)} \mid \theta)\right) + \lambda J(\theta) = \min!$$

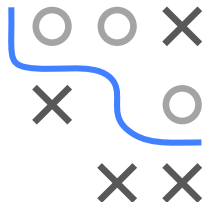


# RISK MINIMIZATION IN ML

- ML models usually consist of the following components:

$$\text{ML} = \underbrace{\text{Hypothesis Space} + \text{Risk} + \text{Regularization}}_{\text{Formulating the optimization problem}} + \underbrace{\text{Optimization}}_{\text{Solving it}}$$

- Hypothesis Space:** Parametrized function space
- Risk:** Measure prediction errors on data with loss  $L$
- Regularization:** Penalize model complexity
- Optimization:** Practically minimize risk over parameter space



### EXAMPLE 3: REGULARIZED LM

- ERM with L2 loss, LM, and L2 regularization term:

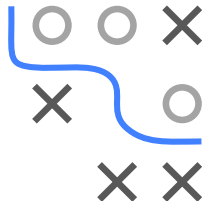
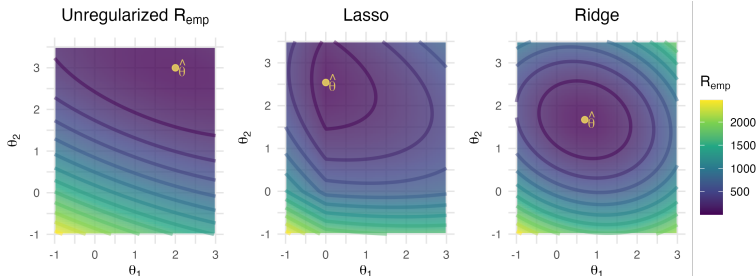
$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \sum_{i=1}^n \left( \boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)} \right)^2 + \lambda \cdot \|\boldsymbol{\theta}\|_2^2 \quad (\text{Ridge regr.})$$

- Problem **multivariate, unconstrained, smooth, convex** and has analytical solution  $\theta = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top \mathbf{y}$ .

- ERM with L2-loss, LM, and L1 regularization:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \sum_{i=1}^n \left( \boldsymbol{\theta}^\top \mathbf{x}^{(i)} - y^{(i)} \right)^2 + \lambda \cdot \|\boldsymbol{\theta}\|_1 \quad (\text{Lasso regr.})$$

- The problem is still **multivariate, unconstrained, convex**, but **not smooth**.

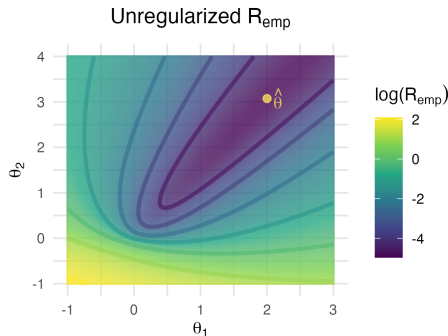
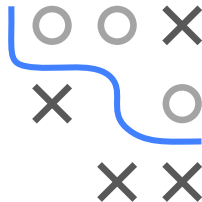


## EXAMPLE 4: (REGULARIZED) LOG. REGRESSION

- For  $y \in \{0, 1\}$  (classification), logistic regression minimizes log / Bernoulli / cross-entropy loss over data

$$\mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^n \left( -y^{(i)} \cdot \theta^\top \mathbf{x}^{(i)} + \log(1 + \exp(\theta^\top \mathbf{x}^{(i)})) \right)$$

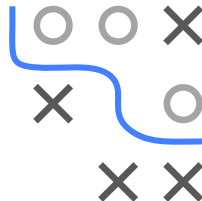
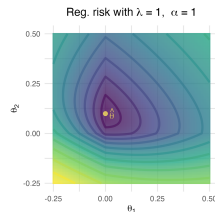
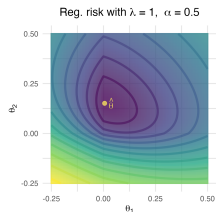
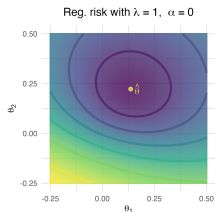
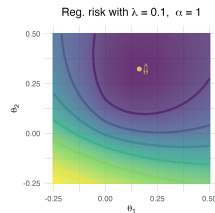
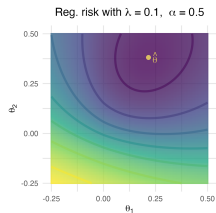
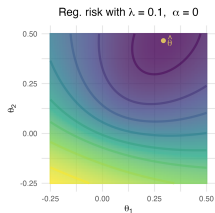
- Multivariate, unconstrained, smooth, convex, not analytically solvable.



# EXAMPLE 4: (REGULARIZED) LOG. REGRESSION

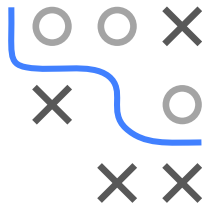
- Elastic net regularization is a combination of L1 and L2 regularization

$$\frac{1}{2n} \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)} | \boldsymbol{\theta})) + \lambda \left[ \frac{1-\alpha}{2} \|\boldsymbol{\theta}\|_2^2 + \alpha \|\boldsymbol{\theta}\|_1 \right], \lambda \geq 0, \alpha \in [0, 1]$$



- The higher  $\lambda$ , the closer to the origin, L1 shrinks coeffs exactly to 0.

## EXAMPLE 4: (REGULARIZED) LOG. REGRESSION



$$\frac{1}{2n} \sum_{i=1}^n L\left(y^{(i)}, f(\mathbf{x}^{(i)} \mid \boldsymbol{\theta})\right) + \lambda \left[ \frac{1-\alpha}{2} \|\boldsymbol{\theta}\|_2^2 + \alpha \|\boldsymbol{\theta}\|_1 \right], \lambda \geq 0, \alpha \in [0, 1]$$

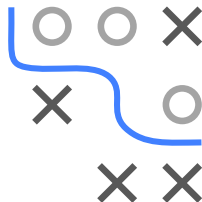
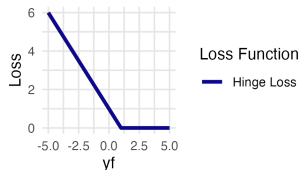
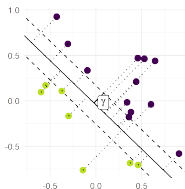
- **Problem characteristics:**

- Multivariate
- Unconstrained
- If  $\alpha = 0$  (Ridge) problem is smooth; not smooth otherwise
- Convex since  $L$  convex and both L1 and L2 norm are convex

# EXAMPLE 5: LINEAR SVM

- $\mathcal{D} = ((\mathbf{x}^{(i)}, y^{(i)}))_{i=1, \dots, n}$  with  $y^{(i)} \in \{-1, 1\}$  (classification)
- $f(\mathbf{x} \mid \theta) = \theta^\top \mathbf{x} \in \mathbb{R}$  scoring classifier: Predict 1 if  $f(\mathbf{x} \mid \theta) > 0$  and  $-1$  otherwise.
- ERM with LM, hinge loss, and L2 regularization:

$$\mathcal{R}_{\text{reg}}(\theta) = \sum_{i=1}^n \max(1 - y^{(i)} f^{(i)}, 0) + \lambda \theta^\top \theta, \quad f^{(i)} := \theta^\top \mathbf{x}^{(i)}$$

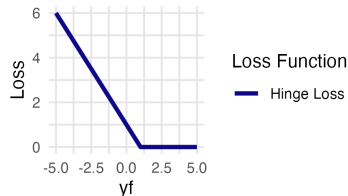
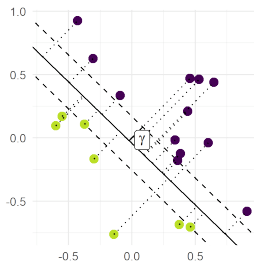
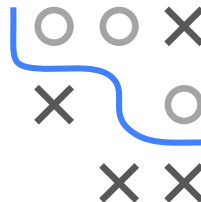


- This is one formulation of the linear SVM.
- Problem is: **multivariate**, **unconstrained**, **convex**, but **not smooth**.

# EXAMPLE 5: LINEAR SVM

- Understanding hinge loss  $L(y, f(\mathbf{x})) = \max(1 - y \cdot f, 0)$

| $y$ | $f(\mathbf{x})$ | Correct pred.? | $L(y, f(\mathbf{x}))$ | Reason for costs        |
|-----|-----------------|----------------|-----------------------|-------------------------|
| 1   | $(-\infty, 0)$  | N              | $(1, \infty)$         | Misclassification       |
| -1  | $(0, \infty)$   | N              | $(1, \infty)$         | Misclassification       |
| 1   | $(0, 1)$        | Y              | $(0, 1)$              | Low confidence / margin |
| -1  | $(-1, 0)$       | Y              | $(0, 1)$              | Low confidence / margin |
| 1   | $(1, \infty)$   | Y              | 0                     | —                       |
| -1  | $(-\infty, -1)$ | Y              | 0                     | —                       |



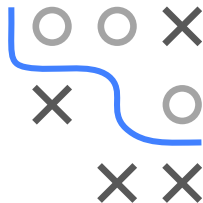
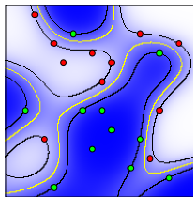
## EXAMPLE 6: KERNELIZED SVM

- **Kernelized** formulation of the primal<sup>(\*)</sup> SVM problem:

$$\min_{\theta} \sum_{i=1}^n L\left(y^{(i)}, \mathbf{K}_i^{\top} \theta\right) + \lambda \theta^{\top} \mathbf{K} \theta$$

with  $k(\cdot, \cdot)$  pos. def. kernel function, and  $\mathbf{K}_{ij} := k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$ ,  $n \times n$  psd kernel matrix,  $\mathbf{K}_i$   $i$ -th column of  $\mathbf{K}$ .

- allows introducing nonlinearity through projection into higher-dim. feature space
- without changing problem characteristics (convexity!)



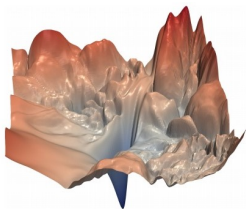
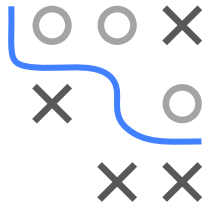
<sup>(\*)</sup> There is also a dual formulation to the problem (comes later!)

# EXAMPLE 6: NEURAL NETWORK

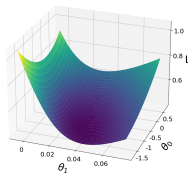
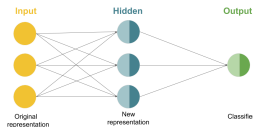
- Normal loss, but complex  $f$  defined as computational feed-forward graph. Complexity of optimization problem

$$\arg \min_{\theta} \mathcal{R}_{\text{reg}}(\theta),$$

- so smoothness (maybe) or convexity (usually no) is influenced by loss, neuron function, depth, regularization, etc.



► [Click for source](#)



Loss landscapes of ML problems. Left: Deep learning model ResNet-56, right: Logistic regression with cross-entropy loss